

BLEND: Balanced and Leaf-Enhanced Dual Fine-Tuning for Taxonomy Completion

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Abstract

Taxonomy completion is the task of integrating new concepts into an existing taxonomy by determining the appropriate hypernym-hyponym relations. Existing approaches often struggle with the inherent imbalance between leaf and non-leaf edges, which induces bias in representation learning. In this paper, we propose **BLEND: Balanced and Leaf-Enhanced Dual Fine-Tuning for Taxonomy Completion**, a novel framework designed to mitigate this inductive bias. Our method employs independent fine-tuning of two lightweight large language models (LLMs): one optimized with a leaf-focused objective and the other trained with a balanced focused strategy. To further enhance structural understanding, we apply contrastive learning over structure-encoded paths and introduce a combined loss function, enabling more robust representation of hierarchical relations. Extensive experiments on three real-world benchmark datasets demonstrate that BLEND achieves up to 9.32% improvement in recall or hit metrics compared to state-of-the-art approaches. Moreover, BLEND delivers efficient inference while outperforming the latest baseline **COMI**, highlighting its effectiveness for taxonomy completion tasks.

CCS Concepts

• Computing methodologies → Information extraction.

Keywords

Dual Fine-Tuning, Taxonomy Completion, Balanced Focused Leaf Focused

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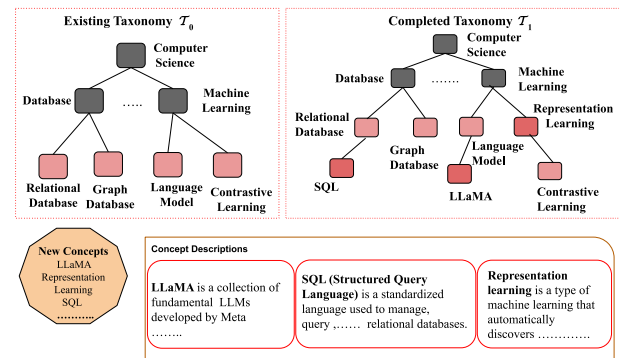


Figure 1: An example of the taxonomy completion task. Given an existing taxonomy T_0 , new concepts such as “LLaMA”, “SQL” and “Representation Learning” are incorporated into the taxonomy.

1 Introduction

A taxonomy is a hierarchical directed acyclic graph (DAG) structured through hypernym-hyponym (“is-a”) relationships among concepts. Taxonomies provide an essential framework for knowledge organization and information extraction [13], enabling efficient indexing and semantic structuring of web data. They play a pivotal role in web-based services such as product search [44], recommendation systems [51], and web content categorization or tagging [15, 26], thereby enhancing search relevance and retrieval quality across web platforms [16, 44].

As new concepts continually emerge, maintaining and expanding existing taxonomies to accommodate them becomes increasingly important. Traditionally, this process relies on domain experts who manually integrate and validate new concepts are both time-consuming and labor-intensive. To address this, recent studies have investigated the task of *Taxonomy Completion (TC)* [1, 10, 37, 39, 50], which aims to automatically insert new concepts (queries) into an existing taxonomy by identifying their most suitable hypernym-hyponym (parent-child) pairs. For example, as illustrated in Figure 1, given a Computer Science taxonomy T_0 , the new concepts “LLaMA” and “Representation Learning” should be positioned under <hypernym (parent), hyponym (child)> pairs such as <“Language

Model, pseudo-leaf”> and <“Machine Learning, Contrastive Learning”> respectively. Here, a *pseudo-leaf* (defined later) represents a virtual node introduced when no hyponym exists in the taxonomy for a given query, serving as a hyponym (child) to determine the most semantically appropriate insertion point.

Prior non-PLM and non-LLM approaches have primarily relied on graph- and structure-based modeling for taxonomy completion. For instance, **TaxoExpan** [30] employs a GNN-based ego-network strategy, while **TaxoEnrich** [10] emphasizes leveraging structural paths within the taxonomy. Similarly, **TMN** [50] integrates multiple semantic scoring mechanisms to improve taxonomy completion performance. In contrast, methods such as **CoSTC** [19, 25, 39, 41] have demonstrated the effectiveness of Pretrained Language Model (PLM)-based encoders [4], which exploit concept descriptions to derive semantically enriched representations—an advantage not available to traditional, non-PLM methods. More recently, **Large Language Models (LLMs)** [18, 52] have exhibited remarkable capabilities in capturing both semantic and structural relationships. Building on this progress, **COMI** [42] leverages fine-tuning **LLaMA-7B** [33] with LoRA adapter [8] to generate semantically and structurally richer concept representations, thereby underscoring the effectiveness of LLM-based representations and mix-up with augmentations techniques [34, 34, 45] for robust learning.

Despite substantial progress, none of the aforementioned studies address the inherent structural bias present in real-world taxonomies. Empirically, existing taxonomies exhibit a pronounced imbalance, with the number of *leaf nodes* (concepts without valid hyponyms) being approximately 2.5–3 times greater than that of *non-leaf nodes*. A leaf node is defined as a parent concept lacking any valid hyponym in the taxonomy. To maintain structural consistency during the taxonomy completion task, we follow prior works [10, 43] and introduce a *pseudo-leaf* node as a virtual hyponym to these actual leaf nodes. In contrast, *non-leaf nodes* are those with valid hyponyms along with the pseudo-leaf node.

This inherent imbalance adversely impacts representation-based learning strategies, as models tend to be biased toward the overrepresented leaf nodes. To address this issue, we propose **BLEND**—a dual fine-tuning framework involving two independently optimized lightweight LLMs. Following **COMI** [42], our approach leverages the semantic and structural richness of LLM-based representations. However, unlike prior work, **BLEND** explicitly mitigates bias by introducing *leaf-focused* and *balanced-focused* fine-tuning strategies, enabling robust and unbiased representation based contrastive learning. Furthermore, we introduce a *structural prompt encoding* mechanism that improves semantics and inference efficiency.

Our main contributions are as follows:

- We introduce **BLEND**, a novel dual fine-tuning framework that employs two lightweight LLMs (each with 1.1B parameters) to explicitly address the inherent imbalance between leaf and non-leaf nodes in taxonomies. This is achieved through two independent *leaf-focused* and *balanced-focused* training strategies.
- We propose a novel **structural prompt encoding** mechanism that jointly models semantic and structural signals along taxonomy paths. Using this encoding for contrastive representation learning with our combined loss formulation, yielding robust and efficient embedding representations.

- Extensive experiments on three real-world benchmark datasets demonstrate the effectiveness of our approach, achieving up to **9.32% improvement** in Recall and Hit metrics over state-of-the-art methods, while maintaining efficient inference performance. The code and trained models are publicly available¹.

2 Related Work

2.1 Taxonomy Expansion and Completion.

Constructing and maintaining a taxonomy from scratch is resource-intensive and demands significant domain expertise. To address this, the *taxonomy expansion* (TE) task was proposed, which integrates emerging concepts as new leaf nodes by linking them to appropriate hypernyms (parent nodes) in an existing taxonomy. Early TE studies such as **HyperExpan** [20] and **TaxoExpan** [30] employed graph neural networks (GNNs) to learn representations over one-hop ego-nets in hyperbolic and Euclidean spaces. Later works, including **BoxTaxo** [11] and **FUSE** [40], adopted box and fuzzy-logic embeddings to capture hierarchical relationships. With the advent of pretrained language models (PLMs) [4, 17, 46, 53], TE methods [19, 47] further advanced via prompt-based learning for semantically enriched concept representation. Motivated by this progress, [50] extended the task to a more practical setting, *taxonomy completion* (TC) task, which aims to insert new concepts as intermediate nodes connecting both parent and child nodes in an existing taxonomy and several works [31, 39, 41, 48] address this.

Recently, large language models (LLMs) have been leveraged for TE [22, 23] and TC [42], demonstrating stronger semantic and structural reasoning. **GenTaxo** [48] and **ICON** [31] generate missing concepts for TC, while **CoSTC** [25] employs inter- and intra-view training with PLM encoders. **Tacoprompt** [41] adopts a cross-encoder design for query–position modeling, though at the cost of high inference latency. To overcome such limitations, **COMI** [42] utilizes LLM-based semantic compression that condenses an entire prompt into a single token [24, 32], enabling efficient LLM inference. Additionally, **COMI** incorporates interpolation-based manifold mixup [35, 36] and cut-based input mixup [9, 49] for robust learning.

None of the existing approaches explicitly address the *inherent structural imbalance* present in taxonomies, where *leaf nodes outnumber non-leaf nodes by nearly a factor of three*, leading to biased representation learning. To mitigate this issue, we propose a **balanced- and leaf-focused dual fine-tuning strategy** involving two independently optimized lightweight LLMs.

2.2 Context Compression in LLMs

Context compression in Large Language Models (LLMs) aims to transform lengthy token sequences into compact implicit representation vectors [6]. This process leverages the model’s internal knowledge and reasoning capabilities to capture essential semantics in a compressed form. Such representations enable LLMs to utilize contextual information more efficiently and significantly reduce memory requirements. For instance, **GraphToken** [27, 32] encodes graph structures into compressed token representations by exploiting graph reasoning abilities. Similarly, **AutoTimes** [18] transforms time-series data into token sequences for autoregressive

¹<https://github.com/Pankaj1881997/BLEND>

predictive modeling. **BLEND** follows this paradigm by compressing task-specific query and novel structural prompt encodings into a single vector representation within the LLM’s latent space for efficient representation-based learning.

2.3 Contrastive representation Learning

Contrastive Learning (CL) is a widely adopted self-supervised learning paradigm that aims to learn discriminative representations by pulling semantically similar samples closer and pushing dissimilar ones apart in the embedding space [28, 38, 44]. It relies on defining positive and negative sample pairs to structure the representation space effectively [14]. In computer vision, methods like SimCLR [3] and MoCo [7] employ data augmentations for this purpose, while in language modeling, SimCSE [5] applies contrastive objectives to improve sentence embeddings. Recent advances such as HCL [6] extend CL to hyperbolic spaces for hierarchical data representation.

Similar to this paradigm, our framework employs a *temperature-scaled binary cross-entropy (BCE) loss* with a positive alignment term. This objective maximizes the similarity between a query and its corresponding positive position, while minimizing the similarity with negative positions within the LLM representation space.

3 Preliminaries & Problem Formulation

In this section, we formally define the taxonomy, the taxonomy completion task and related preliminaries.

Taxonomy: A taxonomy $\mathcal{T}$ is formally defined as a **directed acyclic graph (DAG)** $T = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ denotes the set of nodes (concepts), and $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ denotes the set of directed edges [42]. Each edge $(n_p, n_c) \in \mathcal{E}$ encodes a *hypernym-hyponym* relation, where n_p is a parent (most specific general concept) and n_c is a child (most general specific concept).

Taxonomy Completion: Given a taxonomy $\mathcal{T}_0 = (\mathcal{N}_0, \mathcal{E}_0)$, and a set of new concepts $\mathcal{C}$, the objective of *taxonomy completion* is to integrate $\mathcal{C}$ into $\mathcal{T}_0$. For each new concept $c \in \mathcal{C}$, a suitable parent-child pair (n_p, n_c) is identified, and the edge (n_p, n_c) is replaced by two new edges, namely: (n_p, c) and (c, n_c) . After inserting all query concepts independently, the updated taxonomy is:

$$\mathcal{T}_1 = (\mathcal{N}_0 \cup \mathcal{C}, \mathcal{E}_0 \cup \{(n_p, c), (c, n_c)\}). \quad (1)$$

In the taxonomy completion task, an inherent structural bias arises from the imbalance between leaf and non-leaf nodes. Accordingly, we categorize queries into two distinct types, namely *leaf* and *non-leaf* queries. And the candidate edges set is utilised for sampling during training and for ranking during inference [41, 42].

Candidate Edges: For each node $n \in \mathcal{N}_0$ in the taxonomy $\mathcal{T}_0$, candidate edges are constructed by pairing n with all of its descendant nodes from set $\text{Descendants}(n)$ and denoted as

$$\mathcal{E}_{\text{cand}} = \{(n, m) \mid n \in \mathcal{N}_0, m \in \text{Descendants}(n)\}, \quad (2)$$

Leaf Queries: Queries correspond to parent-child positions (n_p, n_c) where the child n_c is a *pseudo-leaf node* (when no child exists in a taxonomy for a query). To enable such queries, the node set $\mathcal{N}_0$ in the existing taxonomy $\mathcal{T}_0$ is augmented with a pseudo concept as the *pseudo-leaf node*. This node is connected to every node $n \in \mathcal{N}_0$ and is characterized by an empty concept name and description:

$$\text{Name}(n_{\text{leaf}}) = \epsilon, \quad \text{Desc}(n_{\text{leaf}}) = \epsilon, \quad (3)$$

where ϵ denotes the empty string.

Non-Leaf Queries: In a similar manner, queries correspond to parent-child positions (n_p, n_c) , where the child n_c is a valid concept in the taxonomy $\mathcal{T}_0$ but not the pseudo-leaf node.

4 Methodology

In this section, we first introduce how training data is constructed & corresponding queries and prompts are encoded in our decoder-only experimental pipeline. Then, we present the proposed dual LLM-based fine-tuning approach that utilises contrastive learning in the latent space. Furthermore, we also discuss the dual LLM effective edge ranking for inference as illustrated in Figure 2.

4.1 Query and Structural Prompt Encoding

Following prior works, we use a set of nodes $\mathcal{N}_0$ present in the taxonomy as training queries. Each node $q \in \mathcal{N}_0 \setminus \{\text{root}\}$ is treated as a training query concept.

- For every training query q , their corresponding parent-child positions are labeled as **positives**.
- Other edges sampled from $\mathcal{E}_{\text{cand}}$ are treated as **negatives**.

Thus, training consists of alternating positive and negative samples for each training query. Each query concept q is represented using its **name** and **description**, concatenated into a token sequence with a special token [MASK]:

$$\text{QueryTokens}(q) = [\text{Name}(q), \text{Desc}(q), [\text{MASK}]]. \quad (4)$$

For a candidate position (n_p, n_c) , we first sample the longest path from the root node to n_p and subsequently append n_c to this path. To enable effective fine-tuning of LLM, we introduce a *structural prompt encoding* scheme. This encoding constructs a sequence of tokens comprising the names of all nodes along the modified sampled path, together with descriptions of n_p and n_c , as defined below:

$$\text{StructTokens}(n_p, n_c) = [\text{Name}(\text{root}), \dots, \text{Name}(n_p), \text{Desc}(n_p), \text{Name}(n_c), \text{Desc}(n_c), [\text{MASK}]]. \quad (5)$$

Both query and structural tokens are inserted into a prompt template defined below following [42]:

“Summarize the information in MASK token: <tokens>”

These prompts in tokenized form are fed to a **decoder-only LLM**, which compresses it into the single [MASK] token representation, following PromptEOL’s “one-word limitation” principle [6, 29].

4.2 Compressed Hidden Representation based Contrastive Learning

Decoder-only large language models (LLMs) are trained using the causal language modeling (CLM) objective, which optimizes the model to predict the next token conditioned on all preceding tokens in a sequence. Owing to this autoregressive training paradigm and the use of prompt-based conditioning, the hidden state corresponding to the final non-padding token—specifically, the special [MASK] token encapsulates the aggregated semantic representation of the entire input context. Consequently, we extract the final-layer hidden representation of the [MASK] token to serve as the query or structural embedding in our framework. Consider a decoder-only LLM_θ parameterized by θ , where for each query q and parent-child

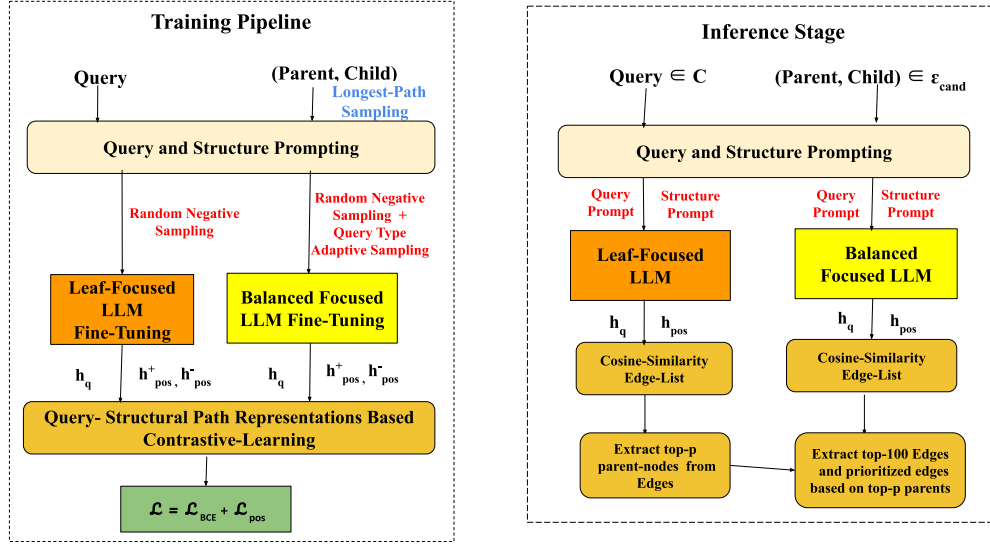


Figure 2: BLEND: a dual independent fine-tuning framework where leaf- and balanced-focused LLMs are combined via inference-time edge prioritization.

pair (n_p, n_c) , the last-layer hidden states h_q and $h_{(n_p, n_c)}$ are denoted as:

$$h_q = LLM_\theta(\text{QueryTokens}(q))_{[\text{MASK}]}^{\text{last}}, \quad (6)$$

$$h_{(n_p, n_c)} = LLM_\theta(\text{StructTokens}(n_p, n_c))_{[\text{MASK}]}^{\text{last}}. \quad (7)$$

We also denote a structural query *more similar* or *less similar* as $(h_{(n_p, n_c)}^+)$ & $(h_{(n_p, n_c)}^-)$, respectively. The final training objective employed is:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{align}}. \quad (8)$$

Here, the binary cross-entropy loss $\mathcal{L}_{\text{BCE}}$ is computed over explicit sets of more similar and less similar pairs:

$$\begin{aligned} \mathcal{L}_{\text{BCE}} = & - \sum_{(q, (n_p, n_c)) \in \mathcal{S}_+} \log \sigma \left(\frac{\text{sim}(\hat{h}_q, \hat{h}_{(n_p, n_c)}^+)}{\tau} \right) \\ & - \sum_{(q, (n_p, n_c)) \in \mathcal{S}_-} \log \left(1 - \sigma \left(\frac{\text{sim}(\hat{h}_q, \hat{h}_{(n_p, n_c)}^-)}{\tau} \right) \right), \end{aligned} \quad (9)$$

$$\text{sim}(\hat{h}_i, \hat{h}_j) = \hat{h}_i^T \hat{h}_j,$$

where $\mathcal{S}_+$ and $\mathcal{S}_-$ denote the sets of more similar and less similar pairs, $\sigma(\cdot)$ is the sigmoid function, and the temperature is set to a small value $\tau \leq 0.01$. Note that all hidden representations are normalized $\hat{h} = \frac{h}{\|h\|_2}$ before computing the cosine similarity or distance as defined in Equation 9. A very small τ sharpens the separation between more and less similar embeddings; however, this also reduces the BCE loss from more similar pairs, providing less signal for the LLM to further align them. To address this, we introduce an additional alignment loss that explicitly minimizes the

distance between queries and more similar structural embeddings:

$$\begin{aligned} \mathcal{L}_{\text{align}} = & \sum_{(q, (n_p, n_c)) \in \mathcal{S}_+} \text{dist}(\hat{h}_q, \hat{h}_{(n_p, n_c)}^+) \\ \text{dist}(\hat{h}_i, \hat{h}_j) = & 1 - \text{sim}(\hat{h}_i, \hat{h}_j) \end{aligned} \quad (10)$$

4.3 Dual LLM Supervised Fine-Tuning

To address the inherent imbalance between leaf and non-leaf queries in taxonomy completion, we adopt a dual fine-tuning framework illustrated in Figure 2. Specifically, two independent LLMs are fine-tuned under contrastive learning objectives but with complementary emphases. This framework enables one model to specialize in leaf-oriented queries, while the other is regularized to maintain balance across leaf and non-leaf queries, thereby mitigating bias.

4.3.1 Leaf-Focused Supervised Fine-Tuning. In the first setting, the LLM is fine-tuned with an explicit emphasis on *leaf-oriented queries*, aiming to strengthen its capability to handle leaf-biased structures within the taxonomy.

- For each query concept $q \in \mathcal{N}_0 \setminus \{\text{root}\}$, we treat q as a leaf query. For non-leaf queries, new positive positions are constructed as (n_p, n_c) , where n_p is the parent of q in $\mathcal{T}_0$ and n_c denotes the pseudo-leaf node.
- The actual positives (n_p, n_c) where $n_c \neq \text{pseudo-leaf}$ are re-defined as negatives, encouraging the model to discriminate pseudo-leaf associations from true hierarchical ones.
- Each resulting query, along with its redefined positives and negatives, is transformed into a prompt (as described in section 4.1) and used to fine-tune the LLM under the loss objective defined in Equation 8.

4.3.2 Balanced-Focused Supervised Fine-Tuning. In the second setting, the complementary LLM is fine-tuned with the goal of maintaining equilibrium between leaf and non-leaf query distributions.

- For every query concept $q \in \mathcal{N}_0 \setminus \{\text{root}\}$, corresponding query and structural prompt encodings are generated for both actual positive and negative positions unlike leaf-focused strategy, following the procedure in section 4.1.
- As leaf nodes typically outnumber internal nodes by approximately a 3:1 ratio, this imbalance introduces bias toward leaf-oriented queries. To counteract this, we apply oversampling of non-leaf queries by duplicating them once, thereby enhancing their representation during training.
- Additionally, we adopt a *query-type adaptive negative sampling* strategy. For leaf queries, s negatives are drawn from non-leaf candidates, while for non-leaf queries, s negatives are sampled from leaf candidates.
- The final negative set comprises both these query-type specific negatives and randomly sampled negatives from $\mathcal{E}_{\text{cand}}$.
- The resulting balanced-focused dataset is then utilized to fine-tune the second LLM under the unified loss objective defined in Equation 8.

4.4 Dual-LLM Effective Edge Ranking for Inference

During inference, the objective is to integrate each new concept $c \in \mathcal{C}$ into the existing taxonomy. To enhance placement accuracy, the leaf-focused LLM is employed in conjunction with the balanced-focused LLM, enabling complementary edge ranking that improves both Hit@1 and Recall@1 metrics for each query concept.

The ranking procedure proceeds in two stages:

- Using the leaf-focused LLM, we rank all candidate edges and select the top-100 edges. From these, we extract their associated parent nodes and retain the top- k parent nodes.
- Using the balanced-focused LLM, we rank candidate edges again. Among the top-100 edges from this model, those whose parent nodes belong to the previously selected top- k parents are prioritized, while the remaining edges are appended in their original order.

Formally, let $\text{Rank}_\ell(q)$ and $\text{Rank}_b(q)$ denote the ranked edge lists for query q produced by the leaf-focused and balanced-focused LLMs, respectively. The final ordered edge set is defined as:

$$\mathcal{R}(q) = \left[\begin{aligned} &\{e \in \text{Top}_{100}(\text{Rank}_b(q)) \mid \text{parent}(e) \in P_q\} \\ &\parallel \{e \in \text{Top}_{100}(\text{Rank}_b(q)) \mid \text{parent}(e) \notin P_q\} \\ &\parallel (\text{Rank}_b(q) \setminus \text{Top}_{100}(\text{Rank}_b(q))) \end{aligned} \right], \quad (11)$$

where P_q denotes the set of top- k ($k = 10$ for all datasets) parents extracted from $\text{Rank}_\ell(q)$, and $\parallel$ indicates list concatenation.

5 Experimental Setup

5.1 Datasets

We evaluate our proposed method on three benchmark taxonomy completion datasets: **SemEval-Food**, **MeSH**, and **WordNet-Verb**, following prior works [37, 41, 42]. The **SemEval-Food** dataset, derived from SemEval-2015 Task 17 [2], corresponds to the food domain. The **MeSH** dataset (**Medical Subject Headings**), obtained as a subgraph of the large-scale biomedical indexing hierarchy [14], represents the clinical domain. The **WordNet-Verb** dataset is constructed from a hierarchy of verb terms from WordNet 3.0, released

Table 1: Overview of dataset characteristics. Here, $|\mathcal{N}|$ and $|\mathcal{E}|$ denote the total number of nodes and edges, respectively. $\mathcal{D}$ represents the maximum taxonomy depth, $\mathcal{T}_{\text{avg}}$ refers to the average token length of node descriptions, and $|\mathcal{E}_{\text{cand}}|$ specifies the number of candidate edges considered for taxonomy completion.

Dataset	$ \mathcal{N} / \mathcal{N}_{\text{train}} $	$ \mathcal{E} $	$\mathcal{D}$	$\mathcal{T}_{\text{avg}}$	$ \mathcal{E}_{\text{cand}} $
SemEval-Food	1486 / 1190	1533	8	34.6	7313
MeSH	9710 / 8072	10498	10	62.6	42970
WordNet-Verb	13936 / 11936	13407	12	26.4	51159

as part of SemEval-2016 Task 14 [12]. For each dataset, we partition 10% of the nodes into the test set $\mathcal{N}_{\text{test}}$, another 10% into the validation set $\mathcal{N}_{\text{validation}}$, and retain the remaining non-overlapping nodes as the training set $\mathcal{N}_{\text{train}}$, consistent with the protocol of [41]. The key statistics of these three benchmark datasets, which highlight their scale, structural complexity, and linguistic properties, are summarized in Table 1.

5.2 Evaluation Metrics

To assess our model’s performance and ensure our results are comparable to previous studies [42], we employ a standard set of ranking-based evaluation metrics:

- **Mean Reciprocal Rank (MRR)** and **Mean Rank (MR)**
- **Hits@k (H@k)** and **Recall@k (R@k)**, where k is set to 1, 5, and 10

Among these, **H@k** and **R@k** are especially informative. They provide a direct measure of placement accuracy by evaluating whether the correct ground-truth position for a new query concept is ranked among the top- k ranked candidates.

5.3 Baseline Methods

We evaluate our approach against several Taxonomy Completion (TC) baselines. **TaxoComplete** [47] leverages query-parent and query-child semantic matching using sentence encoders, where a directionality mechanism enforces hierarchical relations from parent to query and query to child. **TaxoEnrich** [10] employs structural path-based taxonomy contextualized embeddings for query-position matching and incorporates a query-aware sibling aggregator for enhanced robustness. **TMN** [50] learns triplet matching scores across three subtasks: query to position, query to child, and query to parent matching. **CoSTC** [28] introduces inter-view and intra-view negative sampling with a margin based contrastive loss to improve PLM based representation learning. Furthermore, **COMI** [42] utilizes large language models (LLMs) with semantic compression and mix up strategies for robust learning. Following this, we include **CoSTC-LLAMA** as a CoSTC baseline variant that replaces the encoder with an LLAMA based LLM. Among interaction based baselines, we adopt an adapted version of **TEMP** [19], which assigns a single similarity score between query and position path using a PLM encoder, following the setup in [41]. Likewise, **TacoPrompt** [41] models query position interactions by computing three distinct scores using their descriptive texts.

Table 2: Comparison of various taxonomy completion methods across three benchmark datasets. “Total” reports metrics computed over all queries (both leaf and non-leaf), while “Leaf” and “Non-leaf” show metrics for queries whose correct insertion points are leaf or intermediate, respectively. Lower values of MR↓ indicate better performance. Methods marked with † represent interaction-based baselines. The best and second results are in bold and underlined, respectively.

Datasets	Methods	Total								Leaf			Non-leaf		
		MR↓	MRR	R@1	R@5	R@10	H@1	H@5	H@10	MRR	H@5	R@10	MRR	H@5	R@10
SemEval-Food	TaxoExpan	371.29	0.286	5.7	13.3	18.0	11.5	26.4	34.5	0.477	30.1	35.6	0.130	8.0	3.6
	Arborist	256.49	0.290	13.0	18.0	21.0	26.4	34.5	38.5	0.466	39.0	38.5	0.146	12.0	6.7
	TMN	173.52	0.332	10.7	18.7	22.0	21.6	36.5	39.9	0.538	41.5	41.5	0.164	12.0	6.1
	TaxoEnrich	230.42	0.408	11.7	26.7	31.7	23.6	49.3	58.1	0.723	58.5	66.7	0.149	4.0	3.0
	QEN	336.55	0.439	21.9	30.9	35.0	45.9	58.8	64.9	0.732	64.2	68.9	0.209	32.0	9.1
	TaxoComplete	296.07	0.489	14.7	30.0	38.0	29.7	55.4	65.5	0.702	60.2	65.2	0.315	32.0	15.8
	CoSTC	61.47	0.658	18.7	43.0	54.3	39.0	73.4	80.4	0.825	74.5	78.0	0.529	68.0	36.0
	CoSTC-LLaMA	41.46	0.674	22.2	46.0	55.3	46.6	84.5	87.8	0.934	89.4	90.4	0.475	60.0	28.4
	TEMP†	51.37	0.579	20.3	41.2	47.9	42.6	76.4	81.1	0.881	81.3	83.0	0.348	52.0	21.0
	TacoPrompt†	47.42	0.708	30.9	51.1	60.1	64.9	85.8	86.5	0.899	87.8	87.4	<u>0.561</u>	76.0	<u>39.2</u>
	COMI-LLaMA	25.32	<u>0.724</u>	28.9	<u>52.1</u>	<u>61.7</u>	60.8	<u>87.8</u>	93.2	0.945	<u>89.4</u>	92.5	0.555	<u>80.0</u>	38.1
	BLEND	<u>38.05</u>	0.736	<u>29.9</u>	55.3	62.7	<u>62.8</u>	89.8	<u>91.2</u>	<u>0.923</u>	90.2	<u>90.4</u>	0.592	88.0	41.4
MeSH	TaxoExpan	1029.34	0.233	2.7	6.2	12.2	6.0	12.7	23.9	0.381	16.3	24.3	0.137	5.0	4.3
	Arborist	843.20	0.337	5.0	13.6	21.8	11.0	25.8	37.4	0.437	26.7	30.6	0.271	23.8	16.0
	TMN	567.83	0.372	7.2	17.3	24.6	15.9	33.6	43.8	0.525	38.4	40.7	0.271	23.4	14.1
	TaxoEnrich	393.06	0.424	7.4	22.4	31.0	16.2	42.6	52.5	0.619	51.3	54.1	0.296	24.1	15.9
	QEN	451.25	0.438	7.5	21.3	30.8	17.1	43.1	55.9	0.611	51.1	51.8	0.332	26.1	17.9
	TaxoComplete	357.49	0.540	10.8	29.3	41.1	24.5	54.1	63.9	0.605	53.8	52.5	0.500	54.8	34.1
	CoSTC	109.08	0.600	11.0	34.6	47.5	24.9	61.5	72.6	0.741	63.5	66.7	0.512	57.4	35.7
	CoSTC-LLaMA	47.62	0.672	12.9	39.9	54.3	29.4	72.3	82.7	0.822	73.7	74.2	0.579	69.3	41.9
	TEMP†	80.29	0.612	13.8	35.3	48.0	31.4	66.5	77.5	0.839	75.4	77.6	0.471	47.5	29.8
	TacoPrompt†	49.14	0.674	17.9	42.4	55.9	40.7	74.6	84.6	<u>0.868</u>	79.0	81.5	0.554	65.1	40.2
	COMI-LLaMA	29.48	<u>0.727</u>	<u>19.9</u>	<u>47.6</u>	<u>61.5</u>	<u>45.3</u>	<u>79.4</u>	<u>88.5</u>	0.855	80.6	88.7	<u>0.648</u>	<u>76.6</u>	<u>50.3</u>
	BLEND	<u>32.87</u>	0.754	21.0	50.8	64.7	47.8	83.1	89.0	0.876	83.7	<u>82.5</u>	0.68	82.0	53.8
WordNet-Verb	TaxoExpan	1752.27	0.215	4.1	11.4	15.1	6.1	17.1	22.5	0.354	20.5	26.7	0.057	3.1	1.7
	Arborist	1455.25	0.246	3.8	11.0	15.5	5.7	15.5	21.6	0.331	16.2	21.8	0.148	12.8	8.4
	TMN	1513.63	0.290	5.4	14.7	20.7	8.1	21.2	29.1	0.425	23.8	32.8	0.136	10.7	6.8
	TaxoEnrich	5462.08	0.179	3.9	9.0	12.3	5.8	13.6	18.4	0.313	16.8	22.6	0.025	0.5	0.4
	QEN	1730.76	0.404	9.1	23.3	31.0	13.9	34.0	43.9	0.568	38.6	48.4	0.224	15.3	11.8
	TaxoComplete	2661.49	0.407	9.0	22.2	30.9	13.6	31.7	40.8	0.487	32.7	41.3	0.315	27.6	19.1
	CoSTC	241.09	0.505	9.5	27.8	39.1	14.6	39.2	53.1	0.651	41.0	54.7	0.344	31.6	21.8
	CoSTC-LLaMA	<u>176.41</u>	0.545	15.6	34.3	43.1	23.9	51.0	60.5	0.727	55.0	63.6	0.346	34.7	20.5
	TEMP†	960.54	0.450	13.3	30.6	37.5	20.3	45.9	55.0	0.692	53.4	62.8	0.182	15.3	9.5
	TacoPrompt†	436.80	0.557	18.3	36.9	46.5	28.0	52.3	62.5	0.762	56.5	65.8	0.370	35.2	25.3
	COMI-LLaMA	109.45	<u>0.615</u>	<u>19.3</u>	<u>39.9</u>	<u>50.6</u>	<u>29.6</u>	<u>55.5</u>	<u>66.5</u>	<u>0.760</u>	<u>58.5</u>	<u>68.0</u>	<u>0.455</u>	<u>43.4</u>	<u>31.3</u>
	BLEND	233.26	0.618	21.1	42.1	51.2	32.4	58.0	66.6	0.752	61.4	68.3	0.47	43.9	32.4

We also adapt taxonomy expansion baselines **TaxoExpan** [47] and **Arborist** [21] for taxonomy completion tasks by concatenating parent and child node embeddings to construct position representations, following the methodology in [42].

We reproduced baseline models, including TaxoComplete, TaxoEnrich, Temp, and TaxoExpan, using their publicly available GitHub implementations, while the reported results for TacoPrompt, CoSTC, and COMI, were obtained directly from their respective papers. And Arborist and CoSTC-LLaMA results are taken from the COMI paper.

5.4 Implementation Details of BLEND

In our implementation, we employ two lightweight, instruction-tuned variants of *Tiny-LLaMA-1.1B* [33], which are independently fine-tuned for the leaf-focused and balance-focused tasks. Both

models are fully fine-tuned using the AdamW optimizer with a learning rate of 5×10^{-5} and a weight decay of 0.01. Unlike COMI, we do not rely on parameter-efficient tuning methods such as LoRA adapters; instead, we perform full-model fine-tuning to better adapt the LLMs to their respective tasks. All experiments are conducted on NVIDIA A100 GPUs, using either a single 80GB GPU (as in COMI) or two 40GB GPUs for training. The entire training process requires at most 63GB of GPU memory, which is 23GB less than the maximum 86GB required by COMI. This highlights the computational efficiency of our framework, despite employing full fine-tuning in contrast to COMI. During inference, both lightweight LLMs can be efficiently executed independently on a single A100 40GB GPU. For each query, the top-p parent edges from the leaf-focused LLM are stored in JSON format and reused for prioritization.

6 Experimental Results

6.1 Comparison With Baselines

Compared to state-of-the-art methods [41, 42], **BLEND** demonstrates notable improvements in the performance. Specifically, it achieves a **6.14%** gain in H@5 on the SemEval-Food dataset and improvements of **5.52%** and **9.32%** in H@1 for the MeSH and WordNet-Verb datasets, respectively, as reported in Table 2. While the model’s MR on the WordNet-Verb dataset is approximately twice that of the best-performing baseline, its MRR remains competitive. Since MRR is the reciprocal of MR, scaled by a constant factor, it tends to smooth out large deviations caused by a few poorly ranked queries, providing a more balanced measure of performance. In contrast, MR is more sensitive to outliers and easily influenced by higher ranks in the distribution.

Additionally, **BLEND** performs well on non-leaf query concepts, maintaining a high MRR for non-leaf query concepts. Thus, **BLEND** outperforms in Total MRR on all the datasets, with comparable performance in terms of MR. Overall, the consistent improvements across multiple hit and recall metrics highlight the effectiveness of our lightweight bias-minimization strategy. Our approach demonstrates strong effectiveness on non-leaf query concepts, where both parent and child nodes exist within the taxonomy, indicating that it contributes to enhancing the taxonomy’s structural completeness rather than just expansion. For leaf-query concepts, our model performs competitively and, in several cases, surpasses state-of-the-art methods in both recall and hit-based metrics. The improvement is particularly evident in the H@1 metric, which is often considered the good indicator of ranking accuracy. Our gains are more pronounced on the MeSH and WordNet datasets, reflecting the robustness of our approach on large and structurally complex taxonomies, as both datasets contain substantially more nodes and edges than SemEval-Food, which is roughly nine times smaller.

Furthermore, our ablation study, where LLaMA was replaced with BERT, consistently outperforms CoSTC (which also employs a BERT-based encoder) with a 91% improvement in H@1, underscoring the effectiveness of our overall framework. Although our BERT-based pipeline does not outperform another Cross-Encoder BERT-based baseline, TacoPrompt, it is significantly more efficient—TacoPrompt’s inference latency is reported to be up to 284 times higher, as noted in the COMI paper.

6.2 Ablation studies

We conduct a comprehensive ablation study to evaluate the contribution of each component within our proposed framework. In this process, we systematically remove individual components and examine the resulting changes in performance. As summarized in Table 3, we observe a notable performance degradation (with a maximum drop of 12.55%) when the combined loss is replaced with a margin-based loss, similar to that used in prior works [25, 42]. While the margin-based objective aligns with our loss formulation by pulling similar pairs closer and pushing dissimilar ones apart in the latent space, it lacks the integrated optimization behavior achieved by our combined loss. A second major drop (maximum drop of 25%) in performance is observed when the leaf-focused fine-tuned LLM is excluded from the inference stage. In this case,

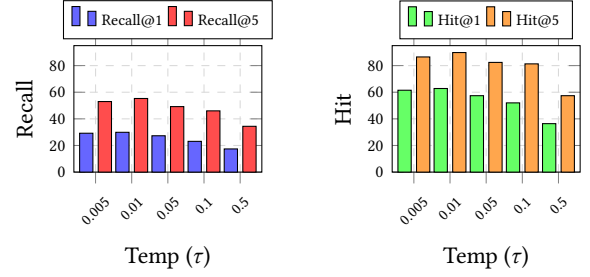
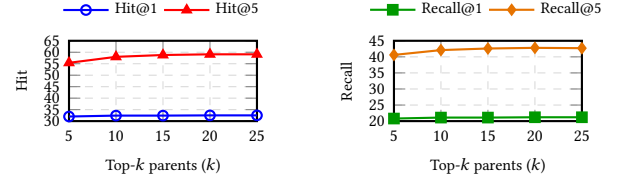


Figure 3: Comparison of Recall and Hit metrics across different temperatures (τ) on SemEval-Food dataset.



(a) Hit metrics across different k values. (b) Recall metrics across different k values.

Figure 4: Analysis of the effect of varying top- k parents extracted from the leaf-focused fine-tuned model on the WordNet-Verb dataset on Hit and Recall metrics.

edge re-ranking relies solely on the balanced-focused model outputs, which limits discrimination capability between positive and negative positions for a new query concept. Moreover, we do not isolate the BCE component from the combined loss in Equation 8, as the alignment term alone cannot act as a self-sufficient objective in contrastive learning, it promotes positive alignment but doesn’t push negatives apart in latent space. Finally, the removal of query-type negative sampling adversely affects the H@1 metric for most datasets (except SemEval-Food), reflecting reduced placement accuracy (H@1). When combined with the exclusion of other loss components, this effect becomes more severe, leading to a consistent performance decline across multiple metrics.

6.3 Further discussions

Our further discussions comprise i) motivation behind using lightweight LLM for taxonomy completion, ii) sensitivity of our proposed framework with respect to the temperature parameter and iii) impact of varying k in top- k while combining outputs from both the LLMs during inference.

Q. Can a lightweight LLM effectively perform structural context compression within a single “MASK” token representation compared to a standard BERT-based encoding?

To investigate this, we replaced the Tiny-LLaMA decoder in BLEND with a BERT encoder [4], while keeping the rest of the pipeline unchanged. The comparison on the MeSH (medical) dataset, presented in Table 4, shows that the lightweight Tiny-LLaMA achieves consistently better performance than BERT. This demonstrates the effectiveness of structural context compression using LLMs for contrastive representation learning in LLM space.

Table 3: Ablation study with datasets as major columns and labeled settings as minor columns. Minor column labels correspond to settings as follows: (a) BLEND, (b) w/o query-type negative sampling, (c) w/o $\mathcal{L}_{align}$, (d) w/o combined loss, (e) w/o query-type negative sampling & $\mathcal{L}_{align}$, (f) w/o Leaf-Focused Tuning. Best result per dataset and metric is bolded.

Metric	Dataset																	
	SemEval-Food						MeSH						WordNet-Verb					
	(a)	(b)	(c)	(d)	(e)	(f)	(a)	(b)	(c)	(d)	(e)	(f)	(a)	(b)	(c)	(d)	(e)	(f)
MRR↑	0.736	0.740	0.737	0.740	0.745	0.708	0.754	0.751	0.758	0.706	0.706	0.729	0.620	0.620	0.620	0.593	0.610	0.610
H@1↑	62.8	62.2	61.4	60.1	61.5	60.8	47.8	43.8	46.5	41.8	41.8	35.4	32.4	29.8	31.2	26.5	29.8	32.0
H@5↑	89.8	89.2	87.1	86.5	87.8	87.8	83.1	80.3	79.5	76.2	76.2	80.1	58.0	58.6	58.5	53.1	58.6	57.7
R@1↑	29.9	29.6	29.2	28.6	29.3	28.9	21.0	19.3	20.5	18.4	18.4	15.6	21.1	19.5	20.4	17.3	19.4	20.8
R@5↑	55.3	54.3	53.0	53.0	53.4	54.0	50.8	47.3	48.8	45.5	45.5	47.8	42.1	42.0	41.8	38.6	41.1	41.6

Table 4: Comparison when Tiny-LLaMA in BLEND is replaced with BERT (denoted as BLEND-BERT) on the MeSH dataset.

Model	MRR↑	R@1↑	R@5↑	H@1↑	H@5↑
BLEND	0.754	21.0	50.8	47.8	83.1
BLEND-BERT	0.680	18.3	42.4	41.0	73.6

Q. How sensitive is our pipeline to the temperature parameter in the temperature-scaled BCE loss? To investigate this, we conducted a detailed sensitivity analysis on the SemEval-Food dataset under varying temperature settings in the temperature-scaled BCE loss used for contrastive learning during balanced LLM fine-tuning. As illustrated in Figure 3, both Recall@1 and Hit@1 metrics exhibit a consistent decline as the temperature increases. This indicates that model performance is highly sensitive to temperature variations, and a lower temperature (particularly $\tau \leq 0.01$) yields more robust discrimination between positives and negatives in the latent space. Empirically, a temperature of $\tau = 0.01$ provides optimal performance.

Q. How does varying the number of top- k parents extracted from the leaf-focused fine-tuned LLM affect the framework’s performance? To analyze this, we evaluated our model on WordNet-Verb dataset by varying k across five different values, retrieving the top- k parents for each case. As illustrated in Figure 4, changes in k have only a marginal effect on both Hit and Recall scores, indicating that the proposed framework is insensitive to the choice of k .

Q. How sensitive is MR on the WordNet-Verb dataset? Table 2 reports MRR for both non-leaf and leaf queries, and since MR is inversely related to MRR, similar trends are expected across both metrics. While our method achieves higher MRR for non-leaf queries, indicating stronger overall ranking quality, performance on leaf queries is lower, suggesting that correct answers tend to appear deeper in the candidate list. Nevertheless, this degradation is moderate, as most positives remain ranked relatively high. However, MR is considerably more sensitive to outliers than MRR, and even a small number of poorly ranked positives can substantially inflate the mean rank. This effect is illustrated by the rank distribution of positives for leaf queries in Table 5: although nearly 90% of positives are ranked within the top 150 candidates, a small fraction

Table 5: Rank distribution of positives among candidate edges for leaf queries on the WordNet-Verb dataset.

Rank	[0, 150)	[150, 200)	[200, 250)	[250, 300)	≥ 300
Percentage(%)	90.55	1.12	1.24	0.62	6.47

(approximately 6.4%) appear beyond rank 300, which has limited impact on MRR but disproportionately increases MR.

Q. How effective is our proposed encoding scheme? To evaluate its effectiveness, we conducted experiments on the MESH dataset both with and without the proposed structural encoding scheme. The results demonstrate that the encoding is crucial: excluding it leads to a **3% reduction in Hit@5** and a **2.16% reduction in Recall@5**, underscoring its role in achieving strong performance.

Q. What is the impact of different weighting schemes in the loss function? We conducted a small-scale study on the SemEval-Food dataset to assess the effect of using different weighting schemes in the loss function. Compared to equal weighting, these alternatives resulted in noticeable performance degradation.

7 Conclusion

In this paper, we present **BLEND**, a dual fine-tuning framework that harnesses the capabilities of lightweight LLMs for structural context compression through our novel structure prompt encoding. The framework also addresses inherent bias in taxonomies, where the number of leaf nodes significantly exceeds that of non-leaf nodes—a factor often overlooked in prior work and one that can lead to biased model behavior. To mitigate this imbalance, we introduce two independent fine-tuning strategies: a *balanced-focused* and a *leaf-focused* approach. These strategies are designed to minimize bias without requiring joint training of multiple LLMs, which is computationally expensive and resource-intensive. The outputs from the leaf-focused fine-tuning are subsequently used to re-rank the edges produced by the balanced-focused model, thereby improving hit and recall performance. Our extensive experiments demonstrate that fine-tuning lightweight LLMs using efficient and targeted strategies can produce results that are competitive with or superior to models that are three to three and a half times larger and fine-tuned with complex joint or mix-up techniques.

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